

CLINDAMYCIN AND METRONIDAZOLE

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CLINDAMYCIN

Lincomycin and clindamycin are the two lincosamide antibiotics currently available in the United States. Lincomycin took its name from Lincoln, Nebraska, where it was first isolated in 1962 from a soil actinomycete, *Streptomyces lincolnensis*. Chemical modification of lincomycin led to the production of clindamycin (1966), which showed better gastrointestinal absorption and a broader antimicrobial spectrum. Because of these advantages, clindamycin has replaced lincomycin in clinical practice. Clindamycin was initially introduced as an antistaphylococcal agent but was later recognized as a potent antianaerobic agent as well.

Structure

Clindamycin is composed of the amino acid trans-1-4-n propylhygrinic acid attached to a sulfur-containing derivative of an octose. It is the 7(S)-chloro-7-deoxy-lincomycin derivative. Its structure is shown in Figure 1.

Pharmacology

Clindamycin in hydrochloride salt or palmitate ester form is absorbed approximately 90% when taken by mouth. Food delays but

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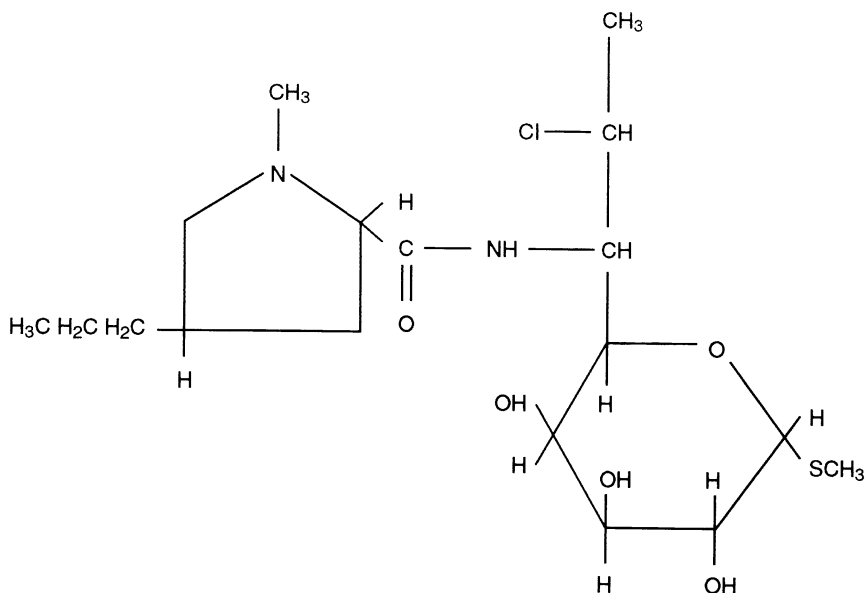


Figure 1. Structure of clindamycin.

does not decrease total absorption. The drug achieves clinically useful concentrations in most tissues.³¹ Notable exceptions are the cerebrospinal fluid even with inflamed meninges and the bile when there is complete biliary obstruction.

Clindamycin is mainly metabolized in the liver, and the drug and its metabolites (such as *N*-demethyl derivative and sulfoxide, which are biologically active) are mostly eliminated in the bile. The amount of clindamycin and its metabolites excreted in urine is about 5% to 10% in adults and 10% to 20% in infants and children. The drug's half-life is 2 to 2.5 hours but is prolonged to 8 to 12 hours in the presence of severe liver disease. Based on these data, dose adjustment should be done for patients with severe liver failure or combined liver and renal failure. No dose adjustment is usually recommended in isolated renal failure, although some investigators suggest dose reduction. In situations such as the above monitoring serum clindamycin level would be ideal but is not practical at the present time. Clindamycin is not removed by hemodialysis or peritoneal dialysis.

Two other points regarding the pharmacokinetics of clindamycin are interesting. First, the drug is actively transported into polymorphonuclear leukocytes and macrophages, significantly increasing the concentration of the drug within these cells.³⁴ Second, studies in animals have shown enterohepatic circulation of clindamycin and its metabolites, leading to prolonged presence of the drug in stool; hence, changes in gut flora may last up to 2 weeks after the discontinuation of clindamycin.²¹

Mechanisms of Activity and Resistance

Clindamycin inhibits bacterial protein synthesis after binding to the 50S ribosomal subunit. The drug affects the process of bacterial peptide chain initiation possibly by interfering with the transpeptidation reaction. The attachment site of clindamycin at the ribosome is the same or overlapping with that of macrolides and chloramphenicol. This explains the antagonism noted *in vitro* between clindamycin and erythromycin; therefore, these antibiotics should not be used in combination.

Clindamycin has been found to potentiate opsonization and phagocytosis of bacteria even at subinhibitory concentrations.¹⁴ By altering the bacterial wall surface as a result of bacterial protein inhibition, clindamycin decreases the adherence of bacteria (e.g., *Staphylococcus aureus*) to host cells and increases intracellular killing of the organism. The compound also inhibits the production of staphylococcal toxin associated with toxic shock syndrome.

The drug is considered a bacteriostatic antibiotic but has demonstrated bactericidal activity for some strains of staphylococci, streptococci, and anaerobes, including *Bacteroides fragilis*, depending on concentration and growth conditions.²⁹ Clindamycin exerts a prolonged postantibiotic effect (i.e., persistent suppression of bacterial growth after short exposure of bacteria to antibiotics) against some susceptible species possibly because of persistence of the drug at the ribosomal binding site.

Bacterial resistance to clindamycin is primarily due to alteration of the target site and does not appear to be related to decreased bacterial drug uptake.⁴¹ Plasmid-mediated transferable resistance has been found in *B. fragilis*. Gram-positive cocci may enzymatically inactivate clindamycin by adenylation, but this is rare and probably not clinically important at least at the present time.

Spectrum of Activity

Clindamycin is active against most anaerobes, most gram-positive cocci, and certain protozoa.

Anaerobes

Clindamycin is active against most gram-positive cocci (*Peptostreptococcus* species, *Peptococcus niger*), gram-positive nonspore-forming bacilli (*Actinomyces* species, *Propionibacterium* species, *Eubacterium*), clostridia (excluding *Clostridium difficile* and a significant percentage of some of the nonperfringens clostridial species, such as *C. tertium*, *C. ramosum*, and *C. sporogenes*), and gram-negative bacilli (*Bacteroides*, *Prevotella*, *Porphyromonas*, and *Fusobacterium* species). Specifically for the clinically important *B. fragilis* and other *B. fragilis* group species, a large survey of isolates from eight hospitals in the United States found that the average rate of resistance to clindamycin was 6.4%.¹⁰

Aerobic Gram-Positive Cocci

Clindamycin is active against streptococci, including β -hemolytic *Streptococcus* group A (*S. pyogenes*), B (*S. agalactiae*), C, and G; *Streptococcus bovis*; microaerophilic streptococci; and most strains of pneumococci and viridans streptococci. The drug is not active against strains of *Enterococcus*.

Clindamycin is generally active against methicillin-susceptible *S. aureus* and *Staphylococcus epidermidis* but usually not active against methicillin-resistant strains; therefore, susceptibility tests should be done when using clindamycin for treatment of staphylococcal infections. During clindamycin treatment of infections caused by clindamycin-susceptible but erythromycin-resistant staphylococci, emergence of clindamycin resistance has been noted in some studies.⁴⁹ Clindamycin is also active against *Corynebacterium diphtheriae* but is relatively inactive against JK group corynebacteria.

Protozoa

Clindamycin (usually in combination with other agents) has good activity against certain protozoal pathogens, such as *Plasmodium* species, *Pneumocystis carinii*, *Toxoplasma gondii*, and *Babesia* species.

Other Microorganisms

Clindamycin is usually active against *Chlamydia trachomatis*. The drug is not active against *Mycoplasma pneumoniae* and *Treponema pallidum*. The drug has poor activity against most species of gram-negative facultative bacteria; exceptions are *Campylobacter* species, including *C. fetus*, *C. jejuni*, and *C. coli*; *Helicobacter pylori*; *Neisseria meningitidis*; and *Neisseria gonorrhoeae*, but clinical efficacy has not been established.

Adverse Effects

The most common adverse effects are diarrhea and hypersensitivity reactions.

Diarrhea

There is no consensus regarding the true incidence of diarrhea with clindamycin treatment, and the reported incidence varies from 2% to 20%. It is clear from the literature and experience that the majority of diarrhea cases during clindamycin treatment represent a mild, self-limited adverse reaction, particularly if the drug is promptly discontinued. The most fearful gastrointestinal event is pseudomembranous colitis caused by overgrowth in stool of *Clostridium difficile*. Pseudomembranous colitis can sometimes be severe, even life-threatening, and susceptibility to acquiring the disease increases with age.

Although clindamycin has been classically associated with pseudomembranous colitis, it is clear that almost all antimicrobials and some cancer chemotherapy agents are potential causes of the condition. Clindamycin, ampicillin or amoxicillin, and cephalosporins are the drugs most frequently implicated.³³ There is great controversy about the incidence of pseudomembranous colitis during clindamycin treatment. The reported incidence ranges from 0.1% to 10%, possibly reflecting differences in the definition of the disease or study methods used but also differences in the true incidence based on characteristics of populations studied (age, underlying disease). In a study of a nosocomial epidemic of *C. difficile* diarrhea, excessive use of clindamycin was implicated, and subsequent restriction of the drug led to control of the epidemic.³²

Pseudomembranous colitis can develop during clindamycin treatment, sometimes quickly, or after the cessation of the drug, sometimes several weeks later. The disease has also been noted after use of topical skin or vaginal clindamycin preparations owing to the small amount of drug absorbed systematically. When severe diarrhea occurs, clindamycin should be discontinued and appropriate management for the initially presumptive diagnosis of *C. difficile* diarrhea started, including avoidance of antimotility agents. Because of the possibility of pseudomembranous colitis, clindamycin should be prescribed with caution, if ever, in patients with inflammatory bowel disease. Besides diarrhea, clindamycin infrequently causes anorexia, nausea, vomiting, flatulence, and a metallic taste.

Hypersensitivity and Allergic Reactions

A generalized mild-to-moderate morbilliform skin rash is reported even more frequently than diarrhea, up to 10% in some series. Urticaria, drug fever, eosinophilia, and erythema multiforme have been reported. Too-rapid intravenous administration of clindamycin has been rarely reported to cause cardiopulmonary arrest and hypotension.

Local Reactions

Induration, pain, and sterile abscess after intramuscular injection of clindamycin; thrombophlebitis after intravenous infusion; and contact dermatitis after topical gel or lotion have all been reported. Symptomatic vaginitis (16%) usually caused by *Candida albicans* and vulvar irritation (6%) are noted after clindamycin vaginal cream.

Rare Adverse Reactions

Transaminases have been noted infrequently to be elevated, sometimes falsely owing to drug interference with the colorimetric measurement. Jaundice and polyarthritis have been reported. Renal and hematopoietic adverse effects have been rarely noted during clindamycin treatment, but direct causal association has not been established. Clinda-

mycin phosphate contains benzyl alcohol as a preservative, and this compound has been associated with a fatal *gasping syndrome* in premature infants.

Drug Interactions

Clindamycin has neuromuscular blocking properties and can potentiate the action of other neuromuscular blocking agents. The drug is physically incompatible and should not be mixed in solutions for parenteral administration with ampicillin sodium, aminophylline, phenytoin sodium, barbiturates, calcium gluconate, and magnesium sulfate.

Clinical Use

Clindamycin has been used in many clinical settings with considerable success.

Intra-Abdominal Infections

Intra-abdominal infections include appendicitis, diverticulitis, perforated ulcer, penetrating trauma, biliary tree infections, infections related to intestinal fistula or ischemic bowel, liver abscess, pancreatic abscess, and other intra-abdominal abscesses. Animal studies and early clinical trials have established the important pathogenic role of anaerobic bacteria in intra-abdominal infections, particularly for abscess formation. Clindamycin has interesting pharmacokinetic properties, such as increased concentration in polymorphonuclear cells and macrophages and less sensitivity to low pH compared with some other antibiotics, that may be important for its activity within abscesses.

The clinical significance of expanded antianaerobic coverage became evident in an early clinical trial that compared cephalothin and kanamycin with clindamycin and kanamycin in patients with penetrating abdominal trauma.⁴⁵ Based on clinical trials and extensive clinical use, the combination of clindamycin with an aminoglycoside became the gold standard for the treatment of intra-abdominal infections with which antibiotics used in most of the subsequent trials were compared. Numerous clinical trials have repeatedly shown the common theme that effective management of intra-abdominal infections in many settings should include broad coverage of both gram-negative facultative rods and anaerobes. Good results have been reported with clindamycin plus an aminoglycoside, clindamycin plus aztreonam, metronidazole plus an aminoglycoside, ampicillin-sulbactam, ticarcillin-clavulanic acid, piperacillin-tazobactam, and imipenem. From the cephalosporins, monotherapy of intra-abdominal infections with agents that have good antianaerobic activity, such as cefoxitin, ceftizoxime, and cefotetan, has been more effective than with agents that are less active against anaerobes.¹⁶ The need to treat enterococci in these mixed anaerobic and aerobic infections

has not been established, and effective treatment of the anaerobes and facultative gram-negative aerobic rods leads to good results.

Pelvic Infections

Pelvic infections include pelvic inflammatory disease, tuboovarian abscess, postpartum endometritis, septic abortion, posthysterectomy vaginal cuff infections, and Bartholin gland abscess. The principles discussed earlier about intra-abdominal infections apply to pelvic infections as well, particularly the need to treat both the aerobic and the anaerobic pathogens. There is an additional pathogenic role for *N. gonorrhoeae* and *C. trachomatis* in pelvic infections. The combination of clindamycin and an aminoglycoside has been shown in many clinical trials to have good results and has been used extensively in clinical practice with success.³⁶

Upper Respiratory Tract Infections

Clindamycin does not have a role in the initial management of acute upper respiratory tract infections. Two circumstances in which clindamycin may be considered, however, occur: First, in cases of chronic sinusitis or chronic otitis when specific bacteriologic data point to anaerobes as contributing or sole etiologic agents, clindamycin may be used. Second, in cases of bacterial pharyngitis that are recurrent or recalcitrant to commonly used regimens, some investigators have suggested that the coexistence of β -lactamase-producing strains of *S. aureus* or *Bacteroides* species with group A streptococci may be the explanation of the failure of agents such as penicillin or erythromycin.⁶ Clindamycin is also an effective third-line agent, after penicillin and erythromycin, in eliminating the carrier state of *Corynebacterium diphtheriae*.

Odontogenic Infections

Orofacial infections may have a dental source and may lead to Ludwig's angina (infection of the floor of the mouth), maxillary sinusitis, retropharyngeal and parapharyngeal abscess, mediastinitis, cavernous sinus thrombosis, and hematogenous dissemination. Penicillin has been traditionally used, but in some although not all¹⁵ trials, clindamycin or cefoxitin had better results.¹⁷ The most likely cause of penicillin failure in these cases was infection with β -lactamase-producing strains of *Prevotella melaninogenica* (formerly *Bacteroides melaninogenicus*).

Pleuropulmonary Infections

Penicillin has been used extensively in pleuropulmonary infections, such as aspiration pneumonia, lung abscess, and empyema, that commonly involve anaerobes. In some²⁶ but not all comparative trials, however, clindamycin led to better results. Overall, for pleuropulmonary

infections that involve anaerobes, clindamycin is a useful alternative in penicillin-allergic patients as well as in patients who do not respond to penicillin and possibly as an appropriate initial treatment in seriously ill patients.³

Diabetic Foot and Decubitus Ulcer Infections

These infections have a mixed bacterial origin that involves aerobic gram-positive cocci and gram-negative rods as well as anaerobes. Antibiotics (single agents or combinations) with the appropriate broad spectrum are necessary for the management of these infections. Clindamycin combined with an agent that has good activity against aerobic gram-negative rods is one of the several regimens used in these infections. Prolonged treatment is often necessary, and clindamycin, having superb bioavailability, is a useful alternative for outpatient management, often combined with a quinolone.

Skin and Soft Tissue Infections

Clindamycin has been used with success in common skin and soft tissue infections, such as cellulitis, furunculosis, carbuncle, folliculitis, impetigo, ecthyma, and hidradenitis suppurativa, as well as in less common but more severe infections, such as necrotizing fasciitis and *Clostridium perfringens* infections. Low-dose oral clindamycin has also been used for prevention of recurrent staphylococcal skin infections.²² For mild skin infections, other antibiotics, usually oral dicloxacillin, are the initially preferred agents. For *C. perfringens*, penicillin is the usually preferred agent. The combination of penicillin and clindamycin, however, produced better results than either drug alone in some animal experiments of *C. perfringens* infections. Clindamycin was superior to penicillin in the treatment of streptococcal myositis in the mouse model.

Acne

Treatment of acne vulgaris is based on the three main pathogenic factors involved: (1) abnormal keratinization of follicular epithelium with the formation of the microcomedone; (2) excess sebum production; and (3) bacteria, mainly *Propionibacterium acnes* and *Propionibacterium granulosum*. Antibacterial treatment with benzoyl peroxide or topical antibiotics such as clindamycin phosphate 1% or erythromycin usually combined with a topical comedolytic agent such as tretinoin has been used.⁴⁶ In more severe forms of acne, antibiotics can be given orally; minocycline, tetracycline, doxycycline, erythromycin, trimethoprim-sulfamethoxazole, and clindamycin have been used. These agents have some important differences in their antibacterial spectrum and specifically activity against *Propionibacterium* species as well as rate and speed of development of resistance, adverse reactions, rates of compliance, and cost. Topical clindamycin has also been used in the treatment of rosacea.

Osteomyelitis

Clindamycin has been used with success in cases of osteomyelitis caused by sensitive strains of *S. aureus* and anaerobes.⁴⁴ The drug has particular value for bone infections related to diabetic foot and decubitus ulcers in which a polymicrobial cause, including anaerobes, is common.

Perioperative Treatment for Head and Neck Surgery

It has been shown that perioperative antibiotics decrease the incidence of wound infections for major head and neck surgery, usually performed for malignancies. Clindamycin combined with gentamicin has been used for this purpose and in some small trials had slightly better results than cefazolin, cefotetan, and ticarcillin-clavulanic acid.

Bacterial Vaginosis

Oral clindamycin and the 2% vaginal cream are alternatives to the more frequently prescribed metronidazole for the treatment of bacterial vaginosis.⁴⁰

Bacillus Species Endophthalmitis

Bacillus species, usually *B. cereus* and *B. subtilis*, are frequently the cause of endophthalmitis related to accidental trauma or bacteremia associated with intravenous drug use. Clindamycin or vancomycin usually combined with an aminoglycoside is the treatment of choice for these infections.

Endocarditis Prophylaxis

Clindamycin is the third-line agent, after amoxicillin and erythromycin, for endocarditis prophylaxis for oral, dental, or upper respiratory tract procedures.¹¹

Toxoplasma Infections

Sulfadiazine with pyrimethamine is currently the first-line treatment for *Toxoplasma* encephalitis, and clindamycin with pyrimethamine²⁷ is the first alternative. In human immunodeficiency virus (HIV)-positive patients, clindamycin treatment is accompanied by higher incidence and severity of adverse reactions, as happens with other drugs in this population. Clindamycin is not recommended for primary or secondary prophylaxis of *Toxoplasma* encephalitis because of high rates of failure and adverse reactions.²⁰ The drug has also been used in cases of ocular toxoplasmosis.

Pneumocystis carinii Infections

Clindamycin in combination with primaquine has been used with good results⁴⁸ in the treatment of *P. carinii* pneumonia in patients with acquired immunodeficiency syndrome (AIDS) and is a useful alternative for patients who are allergic to or cannot tolerate more standard treatment such as sulfamethoxazole-trimethoprim. Clindamycin does not have a role in the primary or secondary prophylaxis of *P. carinii* infections.

Babesiosis

Clindamycin combined with quinine is the current treatment of choice for *Babesia microti* infections. In splenectomized patients with overwhelming infections, exchange transfusion may be necessary.

Malaria

Clindamycin used alone or combined with quinine has been found to have activity against chloroquine-sensitive and chloroquine-resistant strains of *Plasmodium falciparum*.²⁵ Human trials have been limited up to date, however, and more standard regimens are recommended.

Dosage and Route of Administration

Dosage and route of administration should be individualized in all cases in which clindamycin is used, taking into consideration the nature and severity of the infection, the microorganisms involved, possible underlying diseases, level of hepatic and renal function (see earlier pharmacology section), and age of the patient. For some of the indications discussed previously, the recommended oral dose in children is 8 to 16 mg/kg/day for serious infections and 16 to 20 mg/kg/day for more severe infections divided into three or four equal doses. The recommended parenteral dose of clindamycin in neonates (less than 1 month old) is 15 to 20 mg/kg/day divided into three or four equal doses; in children, 20 to 40 mg/kg/day divided into three or four equal doses, with the higher doses used for more severe infections.

For adults, the dosage and route of administration of clindamycin for some clinical uses are shown in Table 1. In a prospective randomized trial, two different doses of clindamycin, combined with gentamicin, were compared in the management of perforated appendicitis. Clindamycin 900 mg intravenously (admixed with gentamicin) every 8 hours produced clinically equivalent results with clindamycin 600 mg intravenously every 6 hours. The first regimen resulted in at least 20% reduction in costs.⁵⁰ In a retrospective study, no differences were found between clinical outcomes in hospitalized patients treated with intravenous clindamycin 600 mg every 6 hours versus 8 hours.⁷

Table 1. CLINICAL USES OF CLINDAMYCIN WITH RECOMMENDED ROUTES OF ADMINISTRATION AND DOSAGE

Use	Route of Administration	Dosage
Susceptible anaerobic infections (often combined with other antibiotics when mixed etiology; see text for specific indications)	IV	1800–2700 mg/day divided into 3 doses (600–900 mg every 8 h)
	oral	150–300 mg 4 times a day; 300–450 mg 4 times a day for more severe infections
Acne rosacea	topical use	1% lotion applied twice a day
<i>Toxoplasma</i> encephalitis	oral	600 mg 4 times a day (combined with pyrimethamine 50–75 mg every day) for 6 weeks for the acute disease. No role in prophylaxis
<i>Pneumocystis carinii</i>	IV/oral	usually 900 mg IV every 8 h initially then orally (combined with primaquine 30 mg orally every day)
Babesiosis	oral	600 mg three times a day (combined with quinine 650 mg) for 7 days
Miscellaneous (bacterial vaginosis, endocarditis prophylaxis, <i>Bacillus</i> species endophthalmitis, malaria)	(see text and quoted references)	

Usage in Pregnancy and Nursing Mothers

Although clindamycin is not contraindicated in pregnancy, in view of the absence of definite extensive human trials, the package insert states that safety for use in pregnancy has not been established. Clindamycin appears in breast milk.

Summary

Clindamycin is an effective antibiotic that has been studied in many clinical trials and used extensively in clinical practice with considerable success. The drug is active against most anaerobes, most gram-positive cocci except enterococci and methicillin-resistant staphylococci, and certain protozoa. Clindamycin has excellent bioavailability and achieves good concentrations in most tissues, excluding the cerebrospinal fluid. The main concern with clindamycin use is *C. difficile* pseudomembranous colitis, which can happen with most antimicrobials, in some instances almost as frequently as with clindamycin.

METRONIDAZOLE

The discovery of azomycin, a nitroimidazole compound isolated from a *Streptomyces* species that had weak activity against *Trichomonas vaginalis*, encouraged active research for synthetic drugs with improved efficacy. These attempts led to the production of metronidazole, which was synthesized in the late 1950s at the Rhone Poulenc Research Laboratories in France. The drug was found initially to be effective against certain protozoan pathogens: *Trichomonas vaginalis*, *Giardia lamblia*, and *Entamoeba histolytica*. In a letter to the editor published in *Lancet* in 1962,³⁷ Shinn reported a patient with *Trichomonas* vaginitis and acute ulcerative gingivitis who had "double cure" after treatment with metronidazole for a week. This clinical observation led to studies that established metronidazole as an important antibiotic for anaerobic infections. The structure of the compound is shown in Figure 2.

Pharmacology

Absorption

Metronidazole is almost completely absorbed when taken by mouth. Food delays but does not decrease total absorption. After the usually given intravenous doses (loading dose 15 mg/kg followed by a maintenance dose of 7.5 mg/kg every 6 hours), peak and trough steady-state serum levels are approximately 25 $\mu\text{g/mL}$ and 18 $\mu\text{g/mL}$. Oral administration of 250- and 500-mg tablets produces peak serum levels of 6 $\mu\text{g/mL}$ and 12 $\mu\text{g/mL}$. The drug's half-life is 8 hours, which means that the dosage interval can be longer than usually recommended. The bioavailability of the compound when given by the rectal route is also good. Metronidazole can also be given intravaginally as 0.75% vaginal gel (20% absorbed systematically) and locally in the 0.75% gel form for skin use.

Distribution and Tissue Penetration

Less than 20% of circulating metronidazole is bound to plasma proteins, and its volume of distribution is 80% of body weight. Metroni-

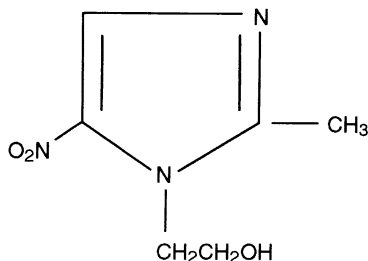


Figure 2. Structure of metronidazole.

dazole has superb tissue penetration in almost all sites, including cerebrospinal fluid, brain abscess contents, saliva, breast milk, and bone. The compound readily crosses the placenta and appears in fetal serum in concentrations similar to those of the mother's serum.

Metabolism and Excretion

Metabolism of metronidazole occurs in the liver with the formation of five main metabolites, including the hydroxy derivative that retains strong antianaerobic activity. The major route of elimination of the parent drug and its metabolites is the urine (60% to 80%), with fecal excretion accounting for a small proportion of the dose. Patients with renal failure may accumulate the hydroxy metabolite of metronidazole, but this has not been associated with any known problems at the present time, so dose reduction is not required²⁸ for treatments of short duration except if there is concomitant liver failure. Dose reduction is recommended in patients with liver failure (usually 50% dose reduction for severe liver failure). Some also suggest dose reduction in severe renal failure. Newborn infants have decreased capability to eliminate metronidazole, and in one study, half-life was inversely related to gestational age. Hemodialysis and peritoneal dialysis remove metronidazole effectively.

Mechanisms of Activity and Resistance

Metronidazole enters cells by passive diffusion and is activated by a reduction process in cells that possess the required enzymatic system, for example, anaerobic bacteria. The nitro group of the drug acts as an electron acceptor donated by ferredoxin-like cell proteins. This reaction decreases the intracellular concentration of the parent drug, creating a gradient that promotes further uptake of metronidazole into susceptible organisms. It is believed that toxic, short-lived derivatives are produced by this mechanism. The exact target of these cytotoxic substances is unclear, but it is probably DNA.

Resistance to metronidazole has been reported rarely for organisms that are considered classically susceptible, such as *T. vaginalis* and species of the *B. fragilis* group. Both slower uptake and decreased intracellular reduction of the drug were found in a strain of *B. fragilis* resistant to metronidazole.⁴² It is likely that decreased intracellular reduction of the drug is the main mechanism of resistance.

Spectrum of Activity

Metronidazole has clinically useful activity against most anaerobes, certain protozoa, and *Helicobacter pylori*.

Anaerobes

Metronidazole has excellent activity against gram-negative anaerobic bacilli (*Bacteroides*, *Prevotella*, *Prophyromonas*, and *Fusobacterium* species), most species of gram-positive anaerobic cocci (*Peptostreptococcus* species, *Peptococcus niger*), and most species and strains of clostridia (including *C. difficile*). Metronidazole does not have good activity against some gram-positive non-spore-forming anaerobic bacilli. *Propionibacterium acnes* is resistant. Approximately 75% of *Actinomyces* and *Arachnia* strains are also resistant. Advantages of metronidazole are its potent bactericidal properties (possibly important for some types of infections, e.g., endocarditis and infections in neutropenic patients) and its activity against nearly all members of the *B. fragilis* group (100% of clinical isolates in some surveys).¹⁰

Protozoa

Metronidazole has good activity against certain protozoa, mainly *T. vaginalis*, *Giardia lamblia*, *Entamoeba histolytica*, and *Balantidium coli*.

Other Microorganisms

Metronidazole is usually active against *H. pylori* and oral spirochetes and variably active against *Campylobacter fetus*, *Gardnerella vaginalis*, and *Treponema pallidum*. The drug is bactericidal against few facultative aerobic gram-negative rods but only when these bacteria are tested under strict anaerobic conditions, so this observation may not have clinical significance. Metronidazole is not active against gram-positive aerobic cocci, including microaerophilic streptococci.

Adverse Effects

Metronidazole is usually well tolerated. The most common adverse reactions are gastrointestinal, and the most serious are related to the nervous system. There has also been a concern about its carcinogenic potential based on studies in experimental animals.

Neurologic Adverse Effects

The most serious reactions are seizures, encephalopathy, and cerebellar dysfunction, which are rare. A more common problem is peripheral neuropathy, occasionally not reversible even with the discontinuation of the drug. This reaction has been mainly noted with prolonged use in patients with Crohn's disease or with high doses. A Swedish group compared three small groups of patients with Crohn's disease: patients who had been receiving metronidazole for at least 1 year with continued use during the study versus patients who had the same long-term treatment but had been off the drug for at least 3 months before

entering the study versus patients who had never received the drug.³⁸ They found no difference in subjective and objective (including neurophysiologic studies) indices of neuropathy between the three groups. It should be noted, however, that the maximum daily dose of metronidazole used was 800 mg and that the power of the study to find statistically significant difference was limited because of the small sample size. In another small study, 6 out of 13 patients with Crohn's disease had symptomatic sensory peripheral neuropathy after 4 to 11 months of treatment with metronidazole.¹² The drug should be given with caution in patients with a history of seizures or other neurologic problems.

Gastrointestinal Adverse Effects

Nausea, epigastric discomfort, anorexia, vomiting, glossitis, stomatitis, furry tongue, and a metallic taste can rarely accompany treatment with metronidazole. Although the drug is used to treat pseudomembranous colitis, it has been reported to be the cause of cases of the disease.³⁵ Rare cases of pancreatitis, improved after discontinuation of the drug, have also been reported.

Interactions with Alcohol, Drugs, and Laboratory Tests

Alcohol should not be taken during metronidazole treatment because a disulfiram-like reaction may occur, including nausea, abdominal cramps, vomiting, headaches, and flushing. Psychotic reactions have been noted in alcoholic patients taking metronidazole and disulfiram concurrently. Metronidazole has been reported to potentiate the anticoagulant effect of warfarin. The drug interferes with certain tests measuring aspartate aminotransferase (AST, SGOT), alanine aminotransferase (ALT, SGPT), lactate dehydrogenase (LDH), and triglycerides.

Miscellaneous Rare Adverse Effects

Reversible mild leukopenia, thrombocytopenia, fever, erythematous rash and pruritus, superficial thrombophlebitis after intravenous use, darkened urine, a sense of pelvic pressure, dysuria, proliferation of *Candida* in the vagina or mouth, dyspareunia, syncope, insomnia, vertigo, weakness, flattening of T wave in electrocardiogram, and fleeting joint pains sometimes resembling *serum sickness* have all been rarely reported.

Concerns About Carcinogenicity and Mutagenicity

Several studies showed evidence of the carcinogenic potential of metronidazole in mice and rats when the drug was given for prolonged periods and in high doses. Prominent among these effects were the development of pulmonary tumors in mice and mammary and hepatic tumors in female rats. No carcinogenicity was found in two studies in

hamsters. Metronidazole showed mutagenic activity in a number of in vitro assay systems, including the Ames Salmonella system, but no definite mutagenicity has been found in mammalian systems.

The above-mentioned studies caused concerns for use of the drug in humans and led to epidemiologic investigations. A cohort of 771 women had long-term follow-up after receiving metronidazole in the 1960s for *T. vaginitis* in Rochester, Minnesota.⁴ The investigators concluded that the analysis of their data "suggests no significant increase in cancer related morbidity or mortality for women exposed to metronidazole for treatment of vaginal trichomoniasis." An increased number of cases of lung cancer, however, was observed in this cohort by using age-specific morbidity and mortality incidence rates from Rochester studies and Cancer Surveillance, Epidemiology, and End-Results Reporting (SEER) data. The limitations of this study are the high proportion of patients lost to follow-up and the significant role of smoking as a confounding factor. It should be noted that these patients were given metronidazole for a short time (10 days). There is more concern in cases in which the drug is given for a long time, such as in patients with Crohn's disease. There are a few reports in the literature of patients with Crohn's disease treated with metronidazole who developed breast and colon cancer, but a cause-and-effect relationship is not established.²⁴

Clinical Use

Metronidazole is used in many settings in current clinical practice.

Anaerobic Infections

Anaerobes have an important pathogenetic role in intra-abdominal infections. Metronidazole, in combination with an antibiotic active against aerobic bacteria, has been used with success in treating these infections. Compared with clindamycin, the theoretic advantages of metronidazole are its potent bactericidal activity and lower potential for *C. difficile* overgrowth. The disadvantages are its more narrow spectrum, specifically the absence of activity against gram-positive cocci such as microaerophilic streptococci. The Canadian Metronidazole-Clindamycin Study Group⁸ conducted a prospective, randomized comparison of metronidazole and clindamycin, each with gentamicin, for the treatment of serious intra-abdominal infections and found no significant difference in efficacy and adverse reactions between the two regimens.

Metronidazole has a particular usefulness in susceptible anaerobic cerebral infections (usually in combination with other antibiotics) and anaerobic meningitis because of its bactericidal activity and good penetration in cerebrospinal fluid and brain, including abscess contents. The drug is also indicated in cases of endocarditis or bacteremia caused by susceptible anaerobic bacteria because of the particular need for potent bactericidal activity in these infections. Metronidazole has also been

used for the treatment of bone, joint, skin and soft tissue, and head and neck infections caused by susceptible anaerobes.

Metronidazole used alone for anaerobic pleuropulmonary infections has led to suboptimal results when these infections had a mixed anaerobic-aerobic cause. The drug is not useful in the majority of cases of actinomycosis and *Propionibacterium* infections.

Crohn's Disease

The cause of Crohn's disease remains unknown, and the management should be individualized for any given patient. The role of metronidazole in the treatment of Crohn's disease has been studied, and trials have produced controversial results.^{2, 30} It seems that the compound is particularly useful for patients with colonic and ileocolonic involvement and in the management of perineal Crohn's disease. Concerns about the safety of long-term treatment with metronidazole temper the enthusiasm for using the drug extensively for this purpose.

Metronidazole has also been used in cases of intestinal bacterial overgrowth, such as in patients with jejunioileal bypass for obesity or dysfunctioning continent ileostomy. A small study found that the compound prevented intrahepatic cholestasis associated with total parenteral nutrition, generating interesting hypotheses for the possible mechanism of this effect.⁹

Clostridium difficile Diarrhea

The high cost of vancomycin and the increase in the incidence of vancomycin-resistant enterococcal infections during the last few years make alternative treatments for the initial management of *C. difficile* diarrhea more desirable. Small trials have shown that metronidazole is effective in the majority of cases.⁴³ A concern has been that the concentration of metronidazole in stool is low because of the excellent absorption of the compound when given by mouth. Although this is probably true for patients with normal transit intestinal time, studies have found bactericidal fecal concentrations of metronidazole in patients who had diarrhea.⁵ In patients who cannot take drugs by mouth or who have ileus or toxic megacolon, metronidazole can be given intravenously. Vancomycin by mouth remains the drug of choice for severe cases of *C. difficile*.

Tetanus

A prospective, open, nonrandomized clinical trial carried out in Indonesia compared metronidazole with procaine penicillin (both as adjuncts to local standard care) in the treatment of moderate tetanus among 173 patients.¹ Doses administered in this trial were metronidazole 500 mg every 6 hours orally or 1 g rectally by suppository every 8 hours versus procaine penicillin 1.5 million units intramuscularly every

8 hours. Patients on metronidazole had a significantly lower mortality rate and a shorter hospital stay.

Helicobacter pylori

The recognition of *H. pylori* in the pathogenesis of gastritis and peptic ulcer disease has been an important advance, which has revolutionized the management of these conditions. Many regimens have been explored in clinical trials.⁴⁷ The main antibiotics used are metronidazole, tetracycline, amoxicillin, and clarithromycin. Other drugs commonly used in trials are bismuth subsalicylate, H₂ blockers, and omeprazole. The first group of trials examined the efficacy, in terms of decreasing ulcer recurrence rates, of bismuth subsalicylate plus antibiotics versus placebo in patients taking H₂ blockers or omeprazole. Because bismuth subsalicylate has both antimicrobial effects on *H. pylori* and direct protective effect on the mucosa, subsequent trials removed this compound from the compared regimens to study the real efficacy of antibiotics.¹⁸ Later the possibility to treat duodenal ulcer with bismuth subsalicylate, metronidazole, and tetracycline without H₂ blockers or omeprazole was studied.¹⁹ Several regimens of antibiotics of different duration have been explored, usually 2 to 4 weeks with good results.

Based on available evidence, it is recommended that *H. pylori*-infected patients with duodenal or gastric ulcers should receive antimicrobial treatment in addition to traditional antiulcer medications as primary treatment for peptic ulcer disease, whether at initial presentation or at the time of recurrence. A frequently used regimen is tetracycline hydrochloride 500 mg four times a day, metronidazole 250 mg three times a day, and bismuth subsalicylate two tablets four times a day for 2 weeks. Amoxicillin 500 mg four times a day can be substituted for tetracycline with only a slight loss in efficacy, and clarithromycin 500 mg three times a day can be substituted for metronidazole whenever metronidazole resistance is suspected. Although laboratory methods for susceptibility testing of *H. pylori* have not been standardized yet, initial studies report that some *H. pylori* strains are resistant to metronidazole.¹³ Several acid pump inhibitor-antimicrobial combinations have been explored also. Omeprazole combined with amoxicillin or clarithromycin has produced good results. The intense basic and clinical research on *H. pylori* infection is expected to clarify some issues in this highly evolving area.

Perioperative Prophylaxis

Perioperative antibiotics can decrease the incidence of infections after colorectal surgery. For elective surgery, an oral regimen (usually erythromycin and neomycin) appears to be more effective than parenteral drugs. Metronidazole has been used also with an oral aminoglycoside as prophylaxis.

Bacterial Vaginosis

The cause of bacterial vaginosis remains unclear. The normal *Lactobacillus*-dominated vaginal flora is replaced by other organisms such as *G. vaginalis*, *Mycoplasma hominis*, *Mobiluncus*, and other anaerobes, but the particular contribution, if any, of these organisms has not been elucidated. Metronidazole 500 mg orally twice a day for 7 days is the usually recommended treatment; trials have found better results with the 7-day regimen compared with single-dose treatment.³⁹ Clindamycin by mouth and topical metronidazole or clindamycin are useful alternatives.

Trichomoniasis

Metronidazole is the current treatment of choice for *Trichomonas* vaginitis. A single oral dose (2 g) is usually as effective as longer regimens. Metronidazole-resistant strains of *T. vaginalis* resulting in treatment failures have been reported.²³

Amebiasis

Metronidazole is the treatment of choice for symptomatic intestinal and hepatic *Entamoeba histolytica* infections. The recommended dose is 750 mg orally three times a day for 10 days. This treatment should be followed by a course of iodoquinol or one of the other intraluminal drugs (paromomycin or diloxanide furoate) to treat asymptomatic intestinal amebiasis.

Giardiasis

Metronidazole 250 mg orally three times a day for 5 days or quina-crine are the usually used drugs for *G. lamblia* infection.

Other Parasites

Metronidazole has also been used in infections caused by *Balantidium coli*, *Entamoeba polecki*, and *Dracunculus medinensis* (guinea worm).

Dosage and Route of Administration

Table 2 lists dosage and route of administration for some common clinical uses of metronidazole. The package insert states that safety and effectiveness in children have not been established except for the treatment of amebiasis.

Table 2. CLINICAL USES OF METRONIDAZOLE WITH ROUTES OF ADMINISTRATION AND DOSAGE

Use	Route of Administration	Dosage
Susceptible anaerobic infections	IV	Loading dose of 15 mg/kg then 7.5 mg/kg every 6 h (usually 1000 mg as loading dose then 500 mg every 6 h)
	oral	1–2 g/day divided in 2 to 4 doses
<i>Clostridium difficile</i> diarrhea	oral	250 mg 4 times a day for 7–10 days
	IV	Loading dose of 15 mg/kg then 7.5 mg/kg every 6 h (IV only for patients with severe ileus)
Nonspecific vaginitis	oral	500 mg twice a day for 7 days
<i>Trichomonas</i> vaginitis	oral	2 g in a single dose or 250 mg 3 times a day for 7 days or 500 mg 4 times a day for 5 days
Amebiasis	oral	750 mg 3 times a day for 10 days
Giardiasis	oral	250 mg 3 times a day for 5 days
Miscellaneous (Crohn's disease, <i>Helicobacter pylori</i> , perioperative prophylaxis, tetanus, <i>Entamoeba polecki</i> , and <i>Dracunculus medinensis</i> [guinea worm])	(see text for discussion, particularly for the evolving role of the drug in the management of Crohn's disease and <i>Helicobacter pylori</i>)	

Usage in Pregnancy and Nursing Mothers

Fetotoxicity studies in experimental animals have given different results depending on the species studied and the dose and route of administration of metronidazole used. The drug should be avoided in pregnancy (particularly during the first trimester) and in nursing mothers.

Other Nitroimidazoles

Although not marketed in the United States, several nitroimidazole compounds, such as tinidazole, ornidazole, nimorazole, and others, have been used extensively in other countries. Tinidazole has a prolonged half-life (12.5 hours) and can be given in once-a-day regimens.

Summary

Metronidazole is a bactericidal antibiotic with strong activity against most anaerobes and certain parasites. The drug has excellent bioavail-

ability and good penetration in most tissues, including the cerebrospinal fluid and brain abscess contents. It is usually well tolerated with few side effects. With a few exceptions, the percentage of anaerobes, including *B. fragilis* group, that are resistant to metronidazole remains low. Metronidazole has been used extensively in clinical practice for years, and its uses have expanded. Its exact role in the management of Crohn's disease and entities related to *H. pylori* is evolving.

References

1. Ahmadsyah I, Salim A: Treatment of tetanus: An open study to compare the efficacy of procaine penicillin and metronidazole. *BMJ* 291:648, 1985
2. Ambrose NS, Allan RN, Kelghley MRB, et al: Antibiotic therapy for treatment in relapse of intestinal Crohn's disease: A prospective randomized study. *Dis Colon Rectum* 28:81, 1985
3. Bartlett JG, Gorbach SL: Penicillin or clindamycin in primary lung abscess? *Ann Intern Med* 98:546, 1983
4. Beard CM, Noller KL, O'Fallon WM, et al: Cancer after exposure to metronidazole. *Mayo Clin Proc* 63:147, 1988
5. Bolton RP, Culshaw MA: Faecal metronidazole concentrations during oral and intravenous therapy for antibiotic associated colitis due to *Clostridium difficile*. *Gut* 27:1169, 1986
6. Brook I, Hirokawa S: Treatment of patients with a history of recurrent tonsillitis due to group A β -hemolytic streptococci. *Clin Pediatr* 24:331, 1985
7. Buchwald D, Soumerai SB, Vandevanter N, et al: Effect of hospital wide change in clindamycin dosing schedule on clinical outcome. *Rev Infect Dis* 11:619, 1989
8. Canadian Metronidazole-Clindamycin Study Group: Prospective, randomized comparison of metronidazole and clindamycin, each with gentamicin, for the treatment of serious intraabdominal infection. *Surgery* 93:221, 1983
9. Capron J-P, Herve M-A, Gineston J-L, et al: Metronidazole in prevention of cholestasis associated with total parenteral nutrition. *Lancet* 1:446, 1983
10. Cuchural GL Jr, Tally FP, Jacobus NV, et al: Susceptibility of the *Bacteroides fragilis* group in the United States: Analysis by site of isolation. *Antimicrob Agents Chemother* 32:717, 1988
11. Dajani AS, Bisno AL, Chung KJ, et al: Prevention of bacterial endocarditis: Recommendations by the American Heart Association. *JAMA* 264:2919, 1990
12. Duffy LF, Daum F, Fisher SE, et al: Peripheral neuropathy in Crohn's disease patients treated with metronidazole. *Gastroenterology* 88:681, 1985
13. European Study Group on Antibiotic Susceptibility of *Helicobacter pylori*: Results of a multicenter survey in 1991 of metronidazole resistance in *Helicobacter pylori*. *Eur J Clin Microbiol Infect Dis* 11:777, 1992
14. Gemmell CG, Peterson PK, Schmeling D, et al: Potentiation of opsonization and phagocytosis of *Streptococcus pyogenes* following growth in the presence of clindamycin. *J Clin Invest* 67:1249, 1981
15. Gilmore WC, Jacobus NV, Gorbach SL, et al: A prospective double-blind evaluation of penicillin versus clindamycin in the treatment of odontogenic infections. *J Oral Maxillofac Surg* 46:1065, 1988
16. Gorbach SL: Antibiotic treatment of anaerobic infections. *Clin Infect Dis* 18(suppl 4):S305, 1994
17. Heimdahl A, von Konow L, Nord CE: Isolation of beta-lactamase-producing *Bacteroides* strains associated with clinical failures with penicillin treatment of human orofacial infections. *Arch Oral Biol* 25:689, 1980
18. Hentschel E, Brandstatter G, Dragosics B, et al: Effect of ranitidine and amoxicillin plus metronidazole on the eradication of *Helicobacter pylori* and the recurrence of duodenal ulcer. *N Engl J Med* 328:308, 1993

19. Hosking SW, Ling TK, Chung SC, et al: Duodenal ulcer healing by eradication of *Helicobacter pylori* without anti-acid treatment: Randomized controlled trial. *Lancet* 343:508, 1994
20. Jacobson MA, Besch CL, Child C, et al: Community programs for clinical research on AIDS. Toxicity of clindamycin as prophylaxis for AIDS-associated toxoplasmic encephalitis. *Lancet* 339:333, 1992
21. Kager L, Liljequist L, Malmborg AS, et al: Effect of clindamycin prophylaxis on the colonic microflora in patients undergoing colorectal surgery. *Antimicrob Agents Chemother* 20:736, 1981
22. Klempner MS, Styrt B: Prevention of recurrent staphylococcal skin infections with low-dose oral clindamycin therapy. *JAMA* 260:2682, 1988
23. Krajden PA, Lossiek JL, Well E, et al: Persistent *Trichomonas vaginalis* infection due to metronidazole resistant strains. *Can Med Assoc J* 134:1373, 1986
24. Krause JR, Ayuyang HQ, Ellis LD: Occurrence of three cases of carcinoma in individuals with Crohn's disease treated with metronidazole. *Am J Gastroenterol* 80:978, 1985
25. Kremsner PG, Winkler S, Brandts C, et al: Curing of chloroquine-resistant malaria with clindamycin. *Am J Trop Med Hyg* 49:650, 1993
26. Levison ME, Mangura CT, Lorber B, et al: Clindamycin compared with penicillin for the treatment of anaerobic lung abscess. *Ann Intern Med* 98:466, 1983
27. Luft BJ, Hafner R, Korzun AH, et al: Toxoplasmic encephalitis in patients with the acquired immunodeficiency syndrome. Members of the ACTG 077/ANRS Study team. *N Engl J Med* 329:995, 1993
28. Mathisen GE, Finegold SM: Metronidazole and other nitroimidazoles. In Gorbach SL, Bartlett JG, Blacklow NR (eds): *Infectious Diseases*. Philadelphia, WB Saunders, 1992, p 260
29. Nastro LJ, Finegold SM: Bactericidal activity of five antimicrobial agents against *Bacteroides fragilis*. *J Infect Dis* 126:104, 1972
30. Palder SB, Shandling B, Bilik R, et al: Perianal complications of pediatric Crohn's disease. *J Pediatr Surg* 26:513, 1991
31. Panzer JD, Brown DC, Epstein WL, et al: Clindamycin levels in various body tissues and fluids. *J Clin Pharmacol* 12:259, 1972
32. Pear SM, Williamson TH, Bettin KM, et al: Decrease in nosocomial *Clostridium difficile*-associated diarrhea by restricting clindamycin use. *Ann Intern Med* 120:272, 1994
33. Pierce PF Jr, Wilson R, Silva J Jr, et al: Antibiotic-associated pseudomembranous colitis: An epidemiologic investigation of a cluster of cases. *J Infect Dis* 145:269, 1982
34. Prokesh RC, Hand WL: Antibiotic entry into human polymorphonuclear leukocytes. *Antimicrob Agents Chemother* 23:373, 1982
35. Saginur R, Hawley CR, Bartlett JG: Colitis associated with metronidazole therapy. *J Infect Dis* 141:772, 1980
36. Sen P, Apuzzo J, Reyelt C, et al: Prospective evaluation of combinations of antimicrobial agents for endometritis after cesarean section. *Surg Gynecol Obstet* 151:89, 1980
37. Shinn DLS: Metronidazole in acute ulcerative gingivitis. *Lancet* 1:1191, 1962
38. Stahlberg D, Barany F, Einarsson K, et al: Neurophysiologic studies of patients with Crohn's disease on long-term treatment with metronidazole. *Scand J Gastroenterol* 26:219, 1991
39. Swedberg J, Steiner JF, Deiss F, et al: Comparison of single dose vs one week course of metronidazole for symptomatic bacterial vaginosis. *JAMA* 254:1046, 1985
40. Sweet RL: New approaches for the treatment of bacterial vaginosis. *Am J Obstet Gynecol* 169:479, 1993
41. Tally FP, Cuchural GL Jr, Malamy MH: Mechanisms of resistance and resistance transfer in anaerobic bacteria: Factors influencing antimicrobial resistance. *Rev Infect Dis* 6(suppl 1):260, 1984
42. Tally FP, Snyderman DR, Shimmel MJ, et al: Mechanisms of antimicrobial resistance of *Bacteroides fragilis*. In Phillips I, Collier J (eds): *Metronidazole: Proceedings of the Second International Symposium on Anaerobic Infections*. Geneva, April 1979. New York, Grune & Stratton, 1979, p 19
43. Teasley DG, Gerding DN, Olson MM, et al: Prospective randomized trial of metronidazole vs vancomycin for *Clostridium difficile*-associated diarrhea and colitis. *Lancet* 2:1043, 1983

44. Templeton WC III, Wawrukiewicz A, Nelo JC, et al: Anaerobic osteomyelitis of long bones. *Rev Infect Dis* 5:692, 1983
45. Thadepalli H, Gorbach SL, Broido PW, et al: Abdominal trauma, anaerobes and antibiotics. *Surg Gynecol Obstet* 137:270, 1973
46. The American Academy of Dermatology: Guidelines of care for acne vulgaris. *J Am Acad Dermatol* 22:676, 1990
47. The Medical Letter on Drugs and Therapeutics: Drugs for treatment of peptic ulcers. *Med Lett* 36:65, 1994
48. Toma E, Fournier S, Dumont M, et al: Clindamycin/primaquine versus trimethoprim-sulfamethoxazole as primary therapy for *Pneumocystis carinii* pneumonia in AIDS: A randomized, double-blind pilot trial. *Clin Infect Dis* 17:178, 1993
49. Watanakunakorn C: Clindamycin therapy of *Staphylococcus aureus* endocarditis. Clinical relapse and development of resistance to clindamycin, lincomycin and erythromycin. *Am J Med* 60:419, 1976
50. Yellin AE, Berne TV, Heseltine PN, et al: Prospective randomized study of two different doses of clindamycin admixed with gentamicin in the management of perforated appendicitis. *Am Surg* 59:248, 1993

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